Chess Endgames on a Cylindrical Chessboard: Exact Combinatorial Game Values, Real-Chess Checks, and Tablebases

Anonymized for review

***Abstract*—Cylinder chess joins the a-file to the h-file, so the board has no edges or corners. We study its king-and-pawn endgames with combinatorial game theory after Elkies: a passed pawn wins, and the kings are frozen in a mutual zugzwang worth zero. We repair this model with a passed-pawn test that respects en passant, and say when the frozen kings are exact. Under the new test and the count rule, an exhaustive census finds no nimber above star with at most six pawns on ordinary boards up to four files wide, but star two appears with seven, in one position and its color-reversed twin, whose values three programs confirm; on the three-file cylinder six pawns suffice, and on seven files ten-pawn rings reach star three. We find king-and-pawn positions worth one-eighth, up-second star and Tiny 1; with frozen kings and queen promotions only, the one-eighth in real chess hinges on one side of one end position. Tablebases confirm that king and rook cannot force mate on the cylinder, where opposite-colored bishops still can. We correct earlier work on cylinder Dawson's chess: the full ring is a first-player win exactly when the strip three files shorter has Grundy value zero.**

***Index Terms*—Chess, combinatorial game theory, endgame tablebases, game solving, nimbers.**

I. Introduction

CYLINDER chess is ordinary chess on a board whose a-file and h-file are joined [1]: a white pawn on h4 can capture on a5, a king on a1 can step to h2, and there are no edges or corners. We use combinatorial game theory (CGT) to study its king-and-pawn endgames.

Under normal play the player who cannot move loses; CGT gives each position a value, and a sum of independent games has the sum of their values [2]–[4]. Chess can be drawn and its kings roam, yet Elkies [5] showed that many pawn endgames split into local fights with exact values, under two conventions:

- a passed pawn wins, and is worth a large number $\pm\mathsf{B}$;
- the kings stand in a mutual zugzwang, the *trébuchet*, worth 0.

The cylinder changes this picture in three ways. There are no rook pawns, and shifting a position sideways does not change its value. A position with a pawn on every file, a *ring*, has no edge at which to cut it. And the corners that decide mates and fortresses are gone.

A nimber $\star k$ is a game in which either player may move to any of $\star 0 = 0$, $\star 1 = \star$, …, $\star(k-1)$. The simplest after $\star = \{\,0 \mid 0\,\}$ is $\star 2 = \{\,0, \star \mid 0, \star\,\}$: either player may move to 0 or to $\star$. Elkies built higher nimbers on large boards from Dawson's pawn game [6], [7], and Drummond-Cole built positions of value $\star 2$ in chess with the help of pieces other than kings and pawns [8]. We ask whether $\star 2$ occurs in a king-and-pawn endgame on an 8-rank board, the $\star 2$ *question*, and whether real chess agrees.

This paper extends an unpublished 2025 study of the basic values [9]: we check every position by computer, correct its errors and carry the analysis on, in the position-by-position format of Wu, Yü and Landry [10]. Theorems and headline numbers carry labels: [P], proved here; [AI], proof given or sketched here (in full in the repository), checked by AI models, human check pending; [C2], two independently written programs agree, the second written from the rules alone; [C1], one program; [Conj], a conjecture. Our contributions are:

- *Model* (Sections II and V): a passed-pawn test that respects en passant, a cut lemma [P], a sum-safety theorem [AI], and a position showing that checking only the top-level components is not enough [C2].
- *Kings* (Section IV): conditions (V1)–(V3) under which a frozen trébuchet gives the result of the whole-board pawn game [AI], and a fourth condition (V4) for the $\pm\mathsf{B}$ value [AI].
- *Small values* (Sections III, VI and VII): cylinder-only values come from rings [P]; $1/8$ and $\uparrow^2\star$ [C2], and Tiny 1 [C1], occur in 10-man positions with a legal pawn history.
- *The $\star 2$ question* (Section VIII): in the default setting (count rule, e.p.-aware test; Section II-D) no $\star 2$ has six pawns or fewer on ordinary boards of width at most four, but $\star 2$ appears with seven [C2]; on the cylinder C_3 a six-pawn ring suffices [C2], and on C_7 some ten-pawn rings are $\star 3$ [C2].
- *Real chess* (Section IX): with the new test the textbook $\star 2$ candidate is not $\star 2$, however draws are treated; with queen promotions only and frozen kings, the $1/8$ hinges on one side of one end position [C1].
- *Tablebases* (Section X): king and rook win only 2.80% of cylinder positions, and on the torus only two major pieces force mate [C2].
- *Dawson's chess* (Section XI): the $3 \times n$ cylinder is a first-player win if and only if the $3 \times (n-3)$ strip has Grundy value 0 [P].
- *Engines* (Section XII): Stockfish is weakest on infinitesimals at low depth, and on sums of components engines often miss the model's winning moves [C1].

Sections II and III set up the rules and the basic values, Sections IV and V add kings and sums, Sections VI–VIII search for small values and $\star 2$, and Section IX tests them against real chess. Sections X–XII turn to tablebases, Dawson's chess

and engines, Section XIII gives the cross-checks and the code, Section XIV lists open problems, and the Appendix gives the FEN of each position with kings.

II. Rules, Values and Conventions

A. The Board and the Moves

We play ordinary chess on the *cylinder*, the 8×8 board with the a-file joined to the h-file. Only the files wrap: ranks 1 and 8 are still edges. Every piece may cross the seam: a white pawn on h4 can capture on a5, a knight on h1 gains a3 and b2, and a king on h1 gains a1 and a2, so a checkmate on the ordinary board need not be one on the cylinder. A sliding piece stops before its ray returns to its own square, so a rook cannot circle a rank as a null move. En passant (e.p.) is allowed, across the seam too. We write C_n and F_n for the cylinder and the ordinary board with n files and eight ranks.

B. Notation

We write $G = \{\, G^L \mid G^R \,\}$ when white (Left) can move to the positions G^L and black (Right) to G^R, and $G + H$ for the sum in which the player to move chooses one component [2]–[4]. Every finite game has a unique *canonical form*, obtained by deleting dominated options and bypassing reversible ones, and $G = H$ exactly when $G - H$ is a second-player win. The *outcome class* of G is $\mathcal{L}$ if white wins whoever starts, $\mathcal{R}$ if black does, $\mathcal{N}$ if the first player wins and $\mathcal{P}$ if the second does. Our values include *numbers* (integers and dyadic fractions such as $1/2 = \{\, 0 \mid 1 \,\}$), *nimbers* $\star k$ (Section I), and *switches* $\pm x = \{\, x \mid -x \,\}$ with $x > 0$. Sums are written without the plus sign, so for a number a, $a \pm x = a + (\pm x) = \{\, a + x \mid a - x \,\}$. The *birthday* of a game is the height of its canonical game tree, so $\star k$ has birthday k. Each game has a *mean* and a *temperature*, read from its thermograph [3]; $\pm x$ has mean 0 and temperature x. The *left* and *right stops* of G are the numbers reached when white or black moves first and both sides play until a number appears; both stops of 0 and of $\star 2$ are 0. Infinitesimals such as $\uparrow = \{\, 0 \mid \star \,\}$ and $\mathrm{tiny}(1) = \{\, 0 \parallel 0 \mid -1 \,\}$ are introduced with their positions in Section III.

C. Normal Play and the Passed-Pawn Convention

Following Elkies [5], we change chess just enough to fit normal play [3].

- In pawn-only positions, a player who has no move loses; there are no null moves.
- A position in which exactly one side has a passed pawn is decided, worth $+\mathsf{B}$ if that pawn is white and $-\mathsf{B}$ if it is black, where B exceeds any count of spare moves (our programs use $\mathsf{B} = 256$).
- Kings are added as a group in mutual zugzwang, the trébuchet, and are then *frozen*: the model never moves them and counts the group as 0, since whoever moves a king there first loses a pawn, and then the game. Section IV asks when this is exact.

A move, usually a capture, can give both sides a passed pawn; such a race needs a rule:

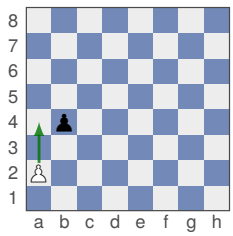


Fig. 1. Position `W a2 / B b4` (ordinary board), value 0. After a2–a4 (arrow) the textbook test calls the a-pawn passed, but b4×a3 en passant removes it; under Definition 1 it is not passed.

- *count*: fewer moves to promote wins; the side that did not move goes first and wins a tie;
- *mover*: the side that made the move wins;
- *opponent*: the other side wins.

None is exact chess, since a new queen may give check or stop the other pawn.

D. En Passant and the Passed-Pawn Test

The textbook test calls a pawn passed if no enemy pawn stands ahead of it on its own file or a neighboring file (on the cylinder, a and h are neighbors). En passant breaks this test: after a2–a4 in Fig. 1 the textbook test calls the a-pawn passed, yet black takes it en passant, b4×a3. Elkies asked for a more systematic treatment of en passant [5]; we use a stricter test.

Definition 1 (E.p.-aware passed pawn): A pawn is *passed* if it passes the textbook test *and* cannot be captured en passant on the opponent's next move.

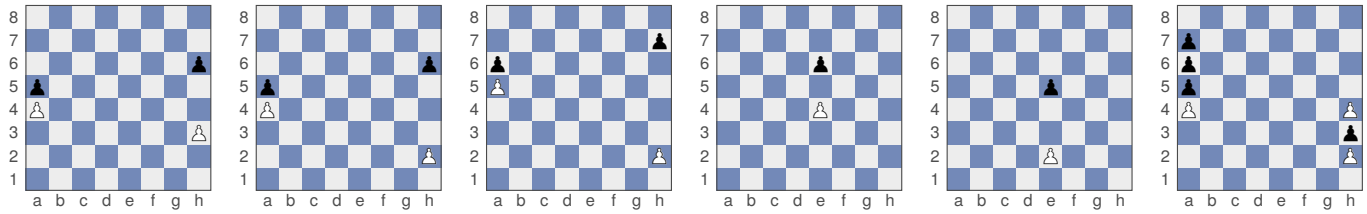
One more white pawn makes the difference decisive. In `W a2 b3 / B b4` black has no move, and the textbook test lets white win at once with a2–a4. Under Definition 1, every white move lets black capture on a3, directly or en passant, and win the race that follows under the count rule, so the value is 0. Across the census of Section VI, every position with at most six pawns on ordinary boards of width at most four, the stricter test changes at least 18,137 of the 231,071 values (a few more changes are hidden where both values exceed the display range), and 4,070 of them in outcome [C1].

The three race rules and three e.p. options (textbook test, e.p.-aware test, e.p. off) give nine *settings*. Every value uses the *default setting*, the count rule with the e.p.-aware test, unless we say otherwise.

E. Components and the Cut Lemma

A pawn changes file only by capturing onto a file where an enemy pawn stands (en passant included), so an empty file stays empty forever and the pawns on either side of it never interact. Each block of neighboring occupied files is a game of its own, a *component*, and a position is the sum of its components; on the cylinder a component is an arc or one *ring* that uses every file.

Theorem 2 (Cut lemma; [P], [C2]): A pawn position on C_n with at least one empty file has the same value as the position on F_{n-1} obtained by cutting the cylinder open along that file.



(a) `W a4 h3 / B a5 h6`: 0 (b) `W a4 h2 / B a5 h6`: 1 (c) `W a5 h2 / B a6 h7`: 2 (d) `W e4 / B e6`: $\star$ (e) `W e2 / B e5`: $\uparrow$ (f) `W a4 h2 h4 / B a5 a6 a7 h3`: $\mathrm{tiny}(1)$

Fig. 2. Pawn positions with the basic values (cylinder: files a and h adjacent). Each value holds in all nine settings (Section II-D) except in (c), which is 2 only with e.p. on and $\Uparrow\star$ with e.p. off.

Proof: No pawn can move onto the empty file, so cutting the board there changes no legal move, no en passant right and no passed-pawn test: the files beside the cut lose only a neighbor that stays empty forever. ∎

Hence values only a cylinder can make must come from rings (Section VI); Section V-C says when a sum of components still names the right winner.

III. Basic Positions on the Cylinder

We now build the basic values of Fig. 2 with pawns alone. Each board leaves a file empty, so by Theorem 2 its value is that of an ordinary-board position, cut open along that file [5], [10]; in four of them the play crosses the seam. A move that the opponent can answer with a passed pawn is a losing move, and it drops out, since having only losing moves is the same as having no moves at all [10]. A move that the opponent can answer by reaching a position at least as good for him as the one we started from is *reversible*: in the canonical form [3] it is replaced by the player's moves from that reply, and so drops out when there are none.

A. The Zero Game

Let us begin with the zero game, in which neither player can move: $0 = \{\,|\,\}$, a second-player win. In `W e4 / B e5` the two pawns are stuck, so the value is 0.

Fig. 2(a) is a zero game that uses the seam. If white goes first, it must play h3–h4, and black captures across the seam, a5×h4, with a passed pawn. If black goes first, h6–h5 is met by a4×h5 in the same way. Whoever moves first loses, so this is again 0.

B. The Games 1 and 2

In Fig. 2(b) the white h-pawn starts one square further back. White can play h2–h3, which leaves the zero game of (a); the double step h2–h4 only loses the pawn to a5×h4. Black has one move, h6–h5, and a4×h5 gives white a passed pawn. Thus $G = \{\,0\,|\,\} = 1$.

In Fig. 2(c) the a-pawns stand one rank higher. White can play h2–h4 or h2–h3, which leave the games (a) and (b) moved up one rank, 0 and 1. Black has only losing moves: h7–h6 is met by a5×h6, and h7–h5 by a5×h6 en passant across the seam. Thus $G = \{\,0, 1\,|\,\} = 2$. With e.p. off, h7–h5 is safe and the value falls to an infinitesimal, $\Uparrow\star$ (defined below). Switching the colors and turning the board through 180° gives -1 and -2 [10].

C. The Infinitesimals $\star$, $\uparrow$ and Tiny 1

Star is the game in which both players can move to 0: $\star = \{\,0\,|\,0\,\}$. In Fig. 2(d), e4–e5 and e6–e5 each lock the file, so $G = \star$, a first-player win.

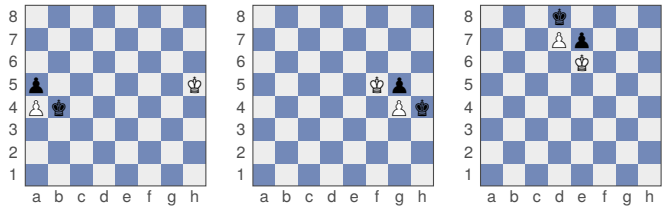
Up is $\uparrow = \{\,0\,|\,\star\,\}$. In Fig. 2(e), white can play e2–e3, which leaves star, or e2–e4, which locks the file: 0. Black can only play e5–e4, which leaves star. Thus $G = \{\,0, \star\,|\,\star\,\}$. Star is reversible for white: black answers e2–e3 with e5–e4, reaching $0 \le G$, since white wins G when black starts. So $G = \{\,0\,|\,\star\,\} = \uparrow$. Up is positive, yet smaller than every positive number: it is an *infinitesimal*. Switching the colors gives $\downarrow = -\uparrow$, and we write $\Uparrow = \uparrow + \uparrow$.

Tiny 1 is $\mathrm{tiny}(1) = \{\,0\,\|\,0\,|\,-1\,\}$, that is, $\{\,0\,|\,\{\,0\,|\,-1\,\}\,\}$. In Fig. 2(f), white can take across the seam, h4×a5, which leaves every pawn blocked: 0. White's h4–h5 drops out, since a6×h5 wins for black. Black can only play a5×h4; from there white locks the a-file with a4–a5, leaving 0, while black plays a6–a5 and keeps the spare move a7–a6, leaving -1. So black's capture leads to $\{\,0\,|\,-1\,\} = -\frac{1}{2} \pm \frac{1}{2}$, and $G = \mathrm{tiny}(1)$, a positive infinitesimal.

A single file can do more. In `W a2 / B a6`, white can play a2–a3, leaving 0 (either move then leaves star), or a2–a4, leaving $\star$. Black can only play a6–a5, which leaves the up position of Fig. 2(e) on the a-file. So $G = \{\,0, \star\,|\,\uparrow\,\}$. As in (e), star is reversible for white, so $G = \{\,0\,|\,\uparrow\,\} = \Uparrow\star$, "double-up star", one of Elkies' examples [5].

IV. Kings: The Trébuchet and When It Is Exact

So far a player with no pawn move simply lost. In chess that player would move the king instead, or, with no king move either, be stalemated, which is a draw. The trébuchet of Section II removes stalemate: a side with no pawn move must move a king, and loses. We now ask when this gives the true chess result. Our king-and-pawn solver plays real chess, with en passant, stalemate and any of the four promotions, but no fifty-move rule; an independent reference solver, written from the rules alone, agrees with it on all 1,418,480 positions of its test runs [C2]. This validates the solver; its results below



(a) W Kh5 a4 / B Kb4 a5 (b) W Kf5 g4 / B Kh4 g5 (c) W Ke6 d7 / B Kd8 e7

Fig. 3. King groups that count as 0: (a) a trébuchet on the cylinder (files a and h adjacent) and (b) the trébuchet T (ordinary board), both real mutual zugzwangs; (c) a king group from the 2025 study (cylinder; Section V), a draw with white to move.

are [C1] unless a second program or the Lichess tablebase [11] confirms them.

A. Two Trébuchets

In Fig. 3(a) only the kings can move. White to move must give up a4, since the only square next to a4 that the king on h5 can move to, h4, is attacked by the a5 pawn across the seam; black to move loses a5 in the same way. Whoever moves first loses a pawn and then the game, so this is a zero game, and with the kings played out a real mutual zugzwang that exists only because of the seam [C2].

Fig. 3(b) shows the same idea on files f to h of the ordinary board; the Lichess tablebase confirms that it is a mutual zugzwang [C2]. We call this group T and place it beside every ordinary-board pawn position in this paper.

B. When Is the Frozen Trébuchet Exact?

A *king group* Z is the two kings and some pawns on a block of files whose neighboring files, the *buffer*, hold no pawns; a *decisive event* is a move that creates a passed pawn, or a move in Z.

Theorem 3 (Validity of the convention; [AI]): Let a position consist of pawn components and a king group Z with its buffer, and suppose that before the first decisive event:

(V1) *the legal chess moves are exactly the moves of the model, nobody is in check, and the side to move has a legal move;*

(V2) *in chess, the side that gets a passed pawn wins, and a race is won as the race rule says;*

(V3) *in chess, every move in Z loses (it is* decisively punished*).*

Then the chess result, with either side to move, is the outcome of $W(X)$, the pawn game that ends at the first passed pawn (Section V-C), without draws, where X is the position with Z and its pawns deleted.

Proof sketch: By (V3) neither side moves in Z without losing, so until the first decisive event both play the pawn game; by (V1) the side to move always has a model move, so play ends at a decisive event, and by (V2) and (V3) that event decides the chess game as it decides $W(X)$. The full proof is in the repository. ■

Corollary 4 [AI]: If also (V4) X is hereditarily sum-safe (Definition 9) and satisfies condition (E) of Theorem 10, and B *exceeds the number of moves either side can make, the chess result in Theorem 3 is the outcome class of the* $\pm\mathsf{B}$ *value of the components.*

Proof: By Theorem 3 the chess result is the outcome of $W(X)$, and by Theorem 10 this is the outcome of $S(X)$. ■

Each condition can fail: a stalemate trap leaves the loser without a move (V1), a passed pawn does not win its race (V2), a king walks over to the pawns or, in Fig. 3(c), white's king moves go unpunished (V3), and a $+\mathsf{B}$ cancels a $-\mathsf{B}$ (V4; Section V-C).

A local test for (V3) is the *certificate of horizon* h: with all other pawns removed, the punisher promotes within h moves of its reply even if the defender may pass at any time, the defender cannot promote anywhere within h moves (the *counter-promotion condition*), and the defender is never stalemated.

Conjecture 5 [Conj]: A certificate of horizon h implies (V3).

Our proof of this has a gap.

Proposition 6 [AI]: On boards with at least six ranks, for $h = 1$, the counter-promotion condition holds in every chess-legal context without a passed pawn.

Proof: A pawn that can promote in one move stands on the rank before its last. With six or more ranks it got there by a single step or a capture, so it cannot be taken en passant, and no enemy pawn can stand ahead of it: it is already passed. ■

Conjecture 7 [Conj]: On a board with room for two files beyond the buffer, for every king group whose punishment needs $h \geq 2$ moves, some context without a passed pawn leaves a move in the group unpunished.

A search we ran once suggests that real trébuchets need $h \approx 5$–7 [Conj], so we trust their zero only beside slow or locked components.

C. Spare King Moves Are Loopy

A king that can step back and forth without loss gives its side unlimited waiting moves, a loopy game [3], [4]: $\mathsf{on} = \{\,\mathsf{on} \mid\,\}$ if only white can wait, $\mathsf{off} = -\mathsf{on}$ if only black can, and dud if both can. Since on exceeds every number, B included, the $\pm\mathsf{B}$ convention fails there, and a decisive event must instead end the game.

Theorem 8 (Outcomes with spare moves; [AI]): Let the components form a short game in which every decisive event ends the game, let a side that can wait pass at any time, and let every other move in Z lose. If only white can wait, black wins exactly when it can force a decisive event in its favor against a white who may pass at any time; otherwise white wins. If both can wait, either side wins exactly when it can do so against a passing opponent; otherwise the game is drawn.

Proof sketch: If only white can wait, every play is finite, since black's moves are, so the game is determined, and black wins exactly when some move of black's wins against every white reply, a pass included. If both can wait, a winning side

never needs to pass, so it wins exactly when it can force a decisive event against passes; otherwise play can go on forever. The full proof is in the repository. ■

Beside a spare king move, then, component values matter only through their forcing threats. The kings alone are already loopy: in normal play (a stalemate loses), every quiet K+P against K placement on the boards from 3×4 to 5×5, summed with sixteen test games, behaves as a decisive win, on, off or dud, or as none of these; none behaves as a number or an infinitesimal, and removing the edges changes the class of 12.5–20.5% of them [C2].

D. How Often Does the Convention Fail?

We solved, with the kings played out, every component with at most two pawns a side beside a trébuchet on boards with six ranks: on one file of the six-file cylinder and of the five-file ordinary board, and on files d and e of the seven-file cylinder (queen promotions only, at most two; on the six-file cylinder the all-promotion results agree in 42 of 42 cases). Of these, 500 start without a race and have a determined result. Under the textbook test, 159 of the 500 get the wrong outcome [C1]: 107 through the kings alone, 18 through en passant and 34 because a passed pawn does not win its race. 143 of the 159 are on the seven-file cylinder. The e.p.-aware test repairs the convention in the 18 e.p. failures, but in 13 of them the free kings change the result again, so the count there falls only from 143 to 138. On the ordinary 8×8 board we checked 94 positions beside a trébuchet against the tablebase. The convention says every one of their 449 king moves loses, yet 41 of them draw and 7 win [C1].

V. Sums, Temperature and the Puzzles Revisited

Most endgames are sums of components. We recall two principles for playing them, apply them to three puzzles, and ask when a sum names the same winner as the whole board.

A. Two Principles

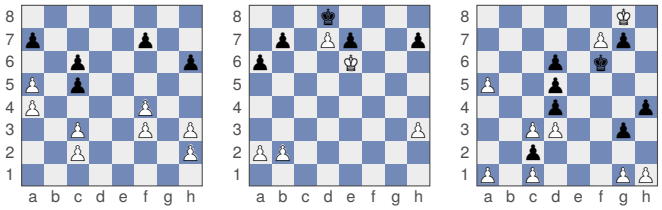
Principle 1 (number avoidance). In a sum of numbers and games that are not numbers, a player who can win can do so without moving in a number [3], [4].

Principle 2 (the hottest move). Move in the component with the highest temperature (Section II-B). For a switch $\{\,x \mid y\,\}$ between numbers $x > y$, the temperature $(x-y)/2$ is what each side gains by moving there first, and infinitesimals have temperature 0. Principle 2 is a heuristic, not a theorem, but it finds the winning move in every example below.

B. The Puzzles Revisited

Fig. 4 shows three puzzles on the cylinder, and Table I compares the values first claimed for them [9] with the exact ones; the earlier walk-throughs pick the right winners, but not always the right values.

Puzzle 1 (Fig. 4(a)) has three components. The earlier values of the c-file, ±1, and of the f-file, $\{\,\{\,2 \mid 1\,\} \mid 0\,\}$ with mean $3/4$, are right. Files h and a are not. White's a5–a6 still wins outright, but black, moving first, plays h6×a5 across the seam



(a) Puzzle 1 (b) Puzzle 2 (c) Puzzle 3

Fig. 4. The three puzzles of [9] (cylinder: files a and h adjacent); Table I gives their exact values. (a) White wins whoever moves first. (b) The pawn part is an infinitesimal. (c) Puzzle 3 as first drawn: $+\mathsf{B}$ (not legal).

TABLE I
Puzzles of Fig. 4: Claimed and Exact Values [C2]

Part	Claimed [9]	Exact
1: c-file	±1	±1
1: f-file	$\{\,\{\,2 \mid 1\,\} \mid 0\,\}$	the same (mean $3/4$)
1: files h, a	1	$\{\,+\mathsf{B} \mid \star\,\}$ (hot)
1: total	$1\pm1+\{\,\{\,2 \mid 1\,\} \mid 0\,\}$	white wins either way
2: pawns	±1	infinitesimal, mean 0
2: kings	0	not a zugzwang
3: files c, d	1	$\mathrm{tiny}(1)$
3: total	2	$+\mathsf{B}$ before any move

and leaves $\star$, not a loss, so this part is $\{\,+\mathsf{B} \mid \star\,\}$, hot, not the number 1. The total is still a white win whoever moves first, but play now starts in the h/a component, by far the hottest (Principle 2). White to move plays a5–a6 and wins. Black to move does best to take on a5 at once, since every other move lets white force a passed pawn (after h6–h5, for example, a4×h5); white then plays c3–c4 in the next hottest component, and $1+\star+\{\,\{\,2 \mid 1\,\} \mid 0\,\}$ is a white win whoever moves next.

Puzzle 2 (Fig. 4(b)) has one pawn component, on files h, a and b. The earlier walk-through finds the right first moves, h3–h4 and a6–a5, and the right outcome, a first-player win. But the value is an infinitesimal with mean 0, not ±1, so by Principle 2 it waits until every hot component is settled. Its king group (Fig. 3(c)) is not a trébuchet either. With black to move, the d7 pawn covers c8 and e8, so Kc7 is forced, and white takes e7 and promotes. With white to move, Kd5, Ke5 and Kf5 all draw: black wins the d7 pawn, but the white king blockades the e-pawn. The group is a zugzwang for black only, on the ordinary board [C2] (our solver and the Lichess tablebase) and on the cylinder [C1], so the real result of this ten-man position is open.

Puzzle 3 (Fig. 4(c)) is illegal, with four white pawns on the first rank, and as drawn it is decided at once, since the pawns on a5 and f7 are passed. On files c and d, white's best move, c3×d4, leaves 0. Black's only move, d4×c3, leaves $\{\,0 \mid -1\,\}$: white locks the files with d3–d4, while black's d5–d4 keeps the spare move d6–d5. Thus $G = \{\,0 \parallel 0 \mid -1\,\} = \mathrm{tiny}(1)$, not 1, and the legal copy one rank up, `W c2 c4 d4 / B c3 d5 d6 d7`, has the same value.

C. When Is the Sum Model Right?

In chess the first passed pawn ends the game; in a sum, a component with a passed pawn becomes $\pm\mathsf{B}$ and play goes on elsewhere, so a $+\mathsf{B}$ can cancel a $-\mathsf{B}$. In $\{\,\mathsf{B} \mid 0\,\}+\{\,0 \mid -\mathsf{B}\,\}$ the first player makes a passed pawn and wins the chess game, but the sum is 0 for every B, since the second player answers in the other component [P]. We write $S(X)$ for the sum model of a pawn position X and $W(X)$ for the whole-board game, which ends at the first passed pawn.

Definition 9 (Hereditary sum safety): A component carries a *threat* for a side if a position reachable from it gives that side a passed pawn. A position X is *hereditarily sum-safe* (HSS) if no position reachable from X by moves of either side, in any order, before the first passed pawn, X included, has two components with threats for opposite sides.

Theorem 10 (Sum safety; [All]): Suppose that X is HSS, that no position reachable from X with two or more components has a usable en passant right (condition (E)), and that B exceeds the number of moves either side can make. Then $S(X)$ and $W(X)$ have the same outcome.

Proof sketch: The models differ only after a passed pawn appears, say for white. By HSS no other component can then give black one, so in the sum black runs out of moves before white's $+\mathsf{B}$ does. The full proof, by induction with (E), is in the repository. ∎

In a random sample of $1{,}500$ positions, all $1{,}297$ HSS ones agree [C1], and with $\mathsf{B} = 256$ the move bound always holds on boards with eight ranks. HSS cannot be dropped (Proposition 11); without (E) the decomposition already fails in the pure pawn game (Proposition 12).

Proposition 11 [C2]: A test of the top-level components alone is not enough: on C_8 with en passant off, $X =$ `W b2 d4 g3 h5 / B a7 c7 f7 g4` is a single component, spanning files f to d across the seam, so such a test passes it; yet $S(X)$ is a black win whoever starts, while $W(X)$ is a first-player win.

Components with opposite threats appear only later in its tree, so X is not HSS.

Proposition 12 [C1]: In the pure pawn game, without the passed-pawn rule (play goes on until a side has no move), (E) cannot be dropped: the sum can be wrong, even in outcome. On the ordinary board with six files and six ranks take `W a4 f4 / B b4 e5`*, where black may take b4×a3 en passant, since white has just played a2–a4. The whole board is $\{\,0 \mid -4, -3 \pm 2\,\}$, a first-player win, while the sum of its two components is $\{\,-1\star \mid -5\star\,\}$, a black win: in the sum the en passant right survives moves in the other component. The two differ with white to move, the case that arises inside larger sums; with black to move, as in the chess position, both say black wins.*

Two models. Our CGT engine adds up components, as Elkies does [5]; the independent whole-board solver ends the game at the first passed pawn. Wherever the two disagree, in every case we tested, our engine with splitting switched off matches the whole-board solver. The two models agree on every headline value: the $1/8$ and $\uparrow^2\star$ positions of Section VII in all nine settings (Section II-D), and the seven-pawn $\star 2$ of Section VIII in the default setting. Elsewhere they can differ, even in outcome: under the opponent rule with the e.p.-aware test, `W a3 b3 c2 e2 / B a6 c5 d5 e3` (the first five-file $\star 2$ of Section VIII-D) is $\Uparrow$ as a sum but $\star 2$ on the whole board. Where they differ, we name the model we mean.

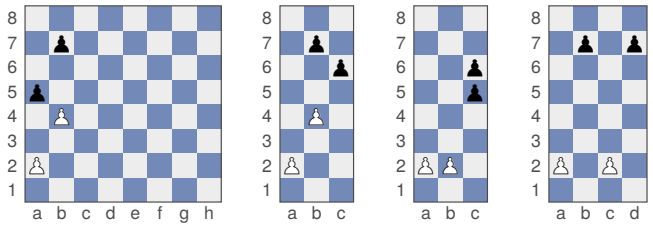


Fig. 5. Atlas values. (a) The hottest, ± 3 (ordinary board). (b)–(d) Values that occur only on cylinders under the textbook test: (b) $3/4$ and (c) $1/4 \pm 1/4$ on C_3 (files a and c adjacent), (d) the $\star 2$ candidate on C_4 (files a and d adjacent).

VI. The Atlas: What Only the Cylinder Can Do

We computed two collections of positions with every pawn on ranks 2–7, at least one pawn per side and no passed pawn. *The atlas* holds every such position with at most five pawns on $C_3, \ldots, C_8$ and $F_3, \ldots, F_8$, up to symmetry: $1{,}223{,}689$ positions [C1]. An independent minimax program confirms their textbook winners for all $12{,}568$ rings and $4{,}000$ random ordinary components. *The census* holds every such position with at most six pawns on $F_1, \ldots, F_4$, counted up to translation and mirror image: $231{,}071$ records standing for $459{,}443$ positions.

Under the textbook test, the atlas values free of B include just two nimbers: $\star$, and one $\star 2$, the textbook candidate of Section VIII, which the e.p.-aware test removes. They also include several ups, such as $\Uparrow\star$ on one file (Section III-C), and fractions with denominator at most 4, of which $3/4$ and $-3/4$ occur only on C_3.

A. The Hottest Position

Apart from $\pm\mathsf{B}$, the highest temperature in the atlas under the textbook test is 3, reached by white a2, b4 against black a5, b7 (Fig. 5(a)). Whoever moves first captures. If white does, b4×a5 leaves the b7-pawn stuck, since either step lets the a5-pawn through, and white three spare moves ahead: 3. If black does, a5×b4 leaves -3. Thus $G = \{\,3 \mid -3\,\} = \pm 3$ [C1].

B. Cylinder-Only Values

A value that occurs on a cylinder but on no ordinary board is *cylinder-only*. By the cut lemma (Theorem 2) it comes from a ring, which on n files needs n pawns.

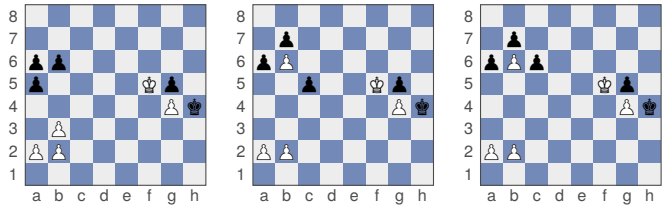
Proposition 13 [C1]: Under the textbook test, 53 atlas values free of B are cylinder-only: 40 from 3-file rings and 13 from 4-file rings. No atlas value occurs only on ordinary boards.

On C_3, white a2, b4 against black b7, c6 is $3/4$ and white a2, b2 against black c5, c6 is $1/4 \pm 1/4$, under both tests

TABLE II
ELKIES' SMALL VALUES IN THE CENSUS

Value	Form	Textbook	E.p.-aware
$1/8$	$\{\,0 \mid 1/4\,\}$	13	9
$1/16$	$\{\,0 \mid 1/8\,\}$	3	2
$1/32$	$\{\,0 \mid 1/16\,\}$	1	0
$\uparrow^2$	$\{\,0 \mid \downarrow\star\,\}$	5	2
$\uparrow^2\star$	$\{\,0, \star \mid \downarrow\,\}$	10	10
$\mathrm{tiny}(1)$	$\{\,0 \parallel 0 \mid -1\,\}$	13	16
$\mathrm{tiny}(x)$, $x \notin \mathbb{Z}$	$\{\,0 \parallel 0 \mid -x\,\}$	0	0

Positions of each exact value in the census (count rule). Textbook column: an independent C solver confirmed each identity (position minus target $= 0$) [C2]. E.p.-aware column: [C1].



(a) (b) (c)

Fig. 6. Elkies' small values beside T (Fig. 3(b)), each a 10-man position with a legal pawn history (ordinary board). (a) $1/8$: `W a2 b2 b3 / B a5 a6 b6`. (b) $\uparrow^2\star$: `W a2 b2 b6 / B a6 b7 c5`. (c) $\mathrm{tiny}(1)$: `W a2 b2 b6 / B a6 b7 c6`.

(Fig. 5(b), (c)). On C_4, white a2, c2 against black b7, d7, the $\star 2$ candidate of Section VIII, is $\{\,3/2 \pm 1/2 \mid -3/2 \pm 1/2\,\}$ under the textbook test but $\{\,2 \pm 1 \mid -2 \pm 1\,\}$ under the e.p.-aware test, a value that also occurs on the ordinary board, so under that test it is no longer cylinder-only.

Nor are these values robust: the 3-file ones depend on the race rule ($3/4$ becomes 0 under the mover rule), the 4-file ones on the e.p. option. With at most four pawns, no value is cylinder-only in all nine settings [C1].

VII. ELKIES' SMALL VALUES

Elkies [5] found fractions, ups and tinies in chess endgames on the ordinary board. We searched the census (Section VI) for smaller values of these kinds: $1/8$, $1/16$ and $1/32$; *up-second* $\uparrow^2 = \{\,0 \mid \downarrow\star\,\}$, positive but smaller than $\uparrow$, and $\uparrow^2\star = \{\,0, \star \mid \downarrow\,\}$; and the *tinies* $\mathrm{tiny}(x) = \{\,0 \parallel 0 \mid -x\,\}$ (a number x). Table II counts them under both passed-pawn tests; under the e.p.-aware test $1/32$ disappears.

A. A Reachable Eighth

Theorem 14 [C2]: In the pawn model, white a2, b2, b3 against black a5, a6, b6 has value exactly $1/8$ in all nine settings (Section II-D) and for every assignment of outcomes to its race positions.

To see why, look at Fig. 6(a). If white moves first, a2–a4 locks the a-file and leaves the zero game. The quieter a2–a3 reaches only $\star$, from which black can return to 0, so it is reversible. The move b3–b4 is bad, since black's reply leaves -3. If black moves first, b6–b5 leaves $1/4$, and a5–a4 is bad. Thus $G = \{\,0 \mid 1/4\,\} = 1/8$.

This pawn structure can arise in a game: from the starting position, c2×b3, a7–a5, b7×a6 and c7×b6 produce it. Its mirror image, white a2, a3, b2 against black a6, b5, b6, cannot, since its a3-pawn could only have come from a2 or b2, and both are still on their starting squares. The independent checks were run on the mirror, which has the same value by symmetry: a C solver confirms both parts of the theorem, and a third, separately written solver confirms the value in all nine settings. Section IX tests the $1/8$ with real kings.

B. Up-Second Star and Tiny 1

White a2, b2, b6 against black a6, b7, c5 (Fig. 6(b)) is $\uparrow^2\star$ in all nine settings [C2]. If white moves first, a2–a4 leaves 0 and a2–a3 leaves $\star$. If black moves first, a6–a5 leaves $\downarrow$. Thus $G = \{\,0, \star \mid \downarrow\,\} = \uparrow^2\star$.

Now move the black c-pawn back to c6 (Fig. 6(c)). If white moves first, a2–a3 leaves 0. If black moves first, a6–a5 leaves $-1/2 \pm 1/2 = \{\,0 \mid -1\,\}$, and c6–c5, which reaches the $\uparrow^2\star$ position, is dominated, since $\{\,0 \mid -1\,\} \le \uparrow^2\star$. Thus $G = \{\,0 \parallel 0 \mid -1\,\} = \mathrm{tiny}(1)$ under every race rule with en passant on (textbook test [C2], e.p.-aware test [C1]); with e.p. off it is $\uparrow$. Unlike the Tiny 1 of Section III-C, it needs no seam. Both pawn structures arise from the starting position by c2×b3, b3–b4–b5–b6, a7–a6 and c7–c5 (c7–c6 for Tiny 1).

C. Two Textbook Values That Fail

Under the textbook test the census also has a $1/8$ at white a2, b2, d2 against black a5, c4, d7 and an $\uparrow^2$ at white a3, c5, d4 against black a7, b7, d7. The e.p.-aware test makes them $-3/2 \pm 3/2$ and $\{\,3/4 \mid \downarrow\star\,\}$. Both rested on counting as passed a pawn that could still be taken en passant. This is why we state Theorem 14 and the $\uparrow^2\star$ above for all nine settings.

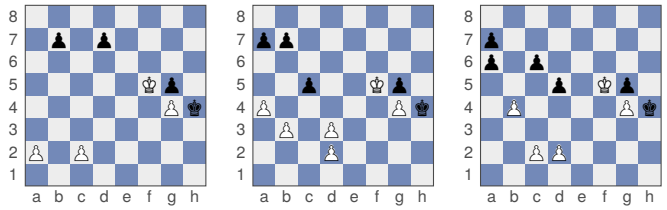
VIII. THE $\star 2$ QUESTION

The $\star 2$ question of Section I asks whether a king-and-pawn position on a board with eight ranks can have this value. We show that the only $\star 2$ in the census under the textbook test comes from that test, that on ordinary boards of width at most four none exists with six pawns or fewer in the default setting, and that $\star 2$ appears with seven.

A. The Textbook Candidate

Fig. 7(a) shows the candidate `W a2 c2 / B b7 d7` on files a–d. Under the count rule with the textbook test, it is the only $\star k$ with $k \ge 2$ in the census of Section VI. If white moves first, a2–a4 leaves $\star$ and c2–c4 leaves 0; a2–a3 and c2–c3 are worse. The position is symmetric (turn the board around and swap the colors), so black's d7–d5 leaves $\star$ and b7–b5 leaves 0. Thus $G = \{\,0, \star \mid 0, \star\,\} = \star 2$.

The weak point is black's plan b7–b5 and b5–b4. After b5–b4 the a-pawn is in the situation of Fig. 1: the textbook test scores a2–a4 as a passed pawn, $+$B, but in chess black takes b4×a3 en passant. Under the e.p.-aware test, `W a2 c2 / B b4 d7` is worth -3, a black win, so b7–b5 is worth -2 ± 1 and, by symmetry, c2–c4 is worth 2 ± 1. The other



(a) Candidate (b) Position A (c) Position B

Fig. 7. Positions of the $\star 2$ question beside T (Fig. 3(b); ordinary board). (a) Candidate: $\star 2$ under the textbook test, $\{\,2 \pm 1 \mid -2 \pm 1\,\}$ under the e.p.-aware test. (b) Position A and (c) position B: $\star 2$ in the default setting (count rule, e.p.-aware test).

moves, among them a2–a4 and d7–d5, which still lead to $\star$, are dominated or reversible. Hence

$$G = \{\,2 \pm 1 \mid -2 \pm 1\,\},$$

a hot game with mean 0 and temperature 2 [C2].

B. No $\star 2$ With Six Pawns or Fewer

Theorem 15 (Census; [C2]): In the census of Section VI, in the default setting, no position has value $\star k$ with $k \geq 2$; the census has 6,724 zeros and 4,042 stars.

Our CGT engine and an independent C solver, which tests whether $G + \star k$ is a second-player win, agree on every record, and the independent whole-board solver finds no $\star k$ with $2 \leq k \leq 292$ among 793,299 labeled placements on widths 1–4 (every placement on each width, without reduction by symmetry or translation). Every position in scope has birthday (Section II-B) at most 292, so no larger nimber can occur. Under the textbook test the census has 6,470 zeros, 3,891 stars and the candidate; the e.p.-aware test removes that $\star 2$. In non-default settings a few other census positions are $\star 2$; Section IX-A tests one of them in real chess.

C. $\star 2$ With Seven Pawns

Theorem 16 ($\star 2$ with seven pawns; [C2]): In the default setting, position A, `W a4 b3 d2 d3 / B a7 b7 c5`, and position B, `W b4 c2 d2 / B a6 a7 c6 d5`, on files a–d of the ordinary board, have value $\star 2$. Neither is $\star 2$ with e.p. off or under the opponent rule.

Both fit beside T, and their pawns can arise in a game, A with the capture c2×d3 and B with b7×a6, pieces taking the other missing men; we checked these histories by hand.

In position A (Fig. 7(b)), if white moves first, a4–a5 leaves 0. The move d3–d4 attacks c5; every black reply except c5×d4 loses at once, and after c5×d4 white continues b3–b4 to $\star$, so the reversible move d3–d4 counts as a move to $\star$. White's third move, b3–b4, loses to c5×b4. If black moves first, a7–a5 leaves 0 and b7–b6 leaves $\star$. Black's other moves are reversible: after a7–a6 and b7–b5 white has replies worth 2 and 3, both at least $\star 2$; and c5–c4 lets d3×c4 free white's d2 pawn for +B. Thus $G = \{\,0, \star \mid 0, \star\,\} = \star 2$.

In position B (Fig. 7(c)), if white moves first, d2–d4 leaves 0 and c2–c3 leaves $\star$. The moves d2–d3 and c2–c4 are reversible, since black's replies reach -2 and -3, and b4–b5 loses to a6×b5. If black moves first, d5–d4 leaves 0, and a6–a5 attacks b4: every white reply except b4×a5 loses at once, and after b4×a5 black continues c6–c5 to $\star$, so a6–a5 counts as a move to $\star$. Black's third move, c6–c5, loses to b4×c5. Thus $G = \star 2$ again. The two analyses match move for move because B is A with the colors reversed and the files mirrored.

TABLE III
VALUES OF THE THREE $\star 2$ POSITIONS IN THE MODEL SETTINGS

Race rule, passed-pawn test	Candidate	A	B
count, e.p.-aware	$\{\,2 \pm 1 \mid -2 \pm 1\,\}$	$\star 2$	$\star 2$
mover, e.p.-aware	$\{\,2 \pm 1 \mid -2 \pm 1\,\}$	$\star 2$	$\star 2$
count or mover, textbook	$\star 2$	$\star 2$	$\star 2$
count or mover, e.p. off	$\star 2$	not $\star 2$	not $\star 2$
opponent, any e.p. option	0	not $\star 2$	not $\star 2$

In Table III (e.p. on unless stated) every entry is [C2]; for A and B, our CGT engine, the census scanner that found them and the independent whole-board solver agree on each. The candidate needs the textbook test but survives e.p. off; A and B need en passant but not the textbook test.

D. Counts and Five Files

On ordinary boards of width at most four, in the default setting, two $\star 2$ positions, A and its color-reversed twin B, have seven pawns and four have eight. One exhaustive scan of 5,503,282 root positions gives these counts, so their completeness is [C1]; our engine re-checked every position it reported [C2]. With Theorem 15, seven is the least number of pawns in a $\star 2$ on these boards [C2].

On five files a scan of all 14,680,983 eight-pawn positions finds 21 of value $\star 2$ in the default setting [C1], each using all five files. As whole-board games all 21 are $\star 2$: the whole-board solver and our engine with splitting switched off agree [C2]. As sums of components (Section V-C) 15 keep the value; the other six change only through the splitting. Three of the 21 come from the 14 eight-pawn $\star 2$ rings on C_5 [C1] by opening them onto files a–e:

- `W a3 b3 c2 e2 / B a6 c5 d5 e3`,
- `W a4 b3 c2 d2 / B a7 b6 c7 e4`,
- `W a6 b4 c4 e3 / B a7 c7 d6 e6`,

and these are $\star 2$ in both models. We call the first *the five-file* $\star 2$. No $\star 3$ or higher nimber occurs in the default setting in any of these scans [C1].

By Theorem 2, a pawn position on C_n with an empty file is one on F_{n-1}, so these scans also cover such positions on C_3 to C_5. A ring has no empty file. One scan of the rings on C_5 to C_8 with at most eight pawns finds $\star 2$ in the default setting only in the 14 eight-pawn rings on C_5 [C1]. Narrower cylinders need fewer pawns. On C_3 the only ring $\star 2$ with six pawns in the default setting is `W a2 a3 b2 / B b7 c6 c7`, and on C_4 the first are `W a4 b2 c2 / B b6 c5 d6 d7` and its color-reversed twin, with seven; our engine confirms both values [C2]. No ring with fewer pawns is a $\star 2$ on either cylinder: the atlas up to five pawns, a C scan of every six- and seven-pawn split, and an independent ring solver agree [C2]. On

C_6 the first rings have nine pawns: a scan of every nine-pawn ring finds six [C1], among them `W a4 c4 d2 e2 / B b6 b7 d6 e5 f6`; both programs confirm three, and the other three are their color-reversed twins [C2]. On C_7 no ring with nine pawns is a $\star 2$, and a third solver written from the rules agrees [C2]; the ten-pawn rings with five pawns a side include 63 of value $\star 2$ [C1], such as `W a2 a3 c3 e3 f2 / B a4 b5 d5 f5 g5`, which all three programs confirm [C2]. So the first ring $\star 2$ on C_n has $n+3$ pawns for $n = 3, \ldots, 7$ [C1]. The C_3 ring is a $\star 2$ with one pawn fewer than any ordinary board of width at most four allows, and it needs two en passant captures: without black's b4×a3 it is $\uparrow$, without white's b5×c6 it is $\downarrow$, and with e.p. off it is 0 [C2].

A $\star 3$ lies just beyond these scans, since $\star 2 + \star = \star 3$. In the default setting, position A beside a one-file group worth $\star$, `W a4 b3 d2 d3 f4 / B a7 b7 c5 f6`, is $\star 3$ on six files with nine pawns, and our engine as a sum, our engine as a whole-board game and the independent whole-board solver agree [C2]. With e.p. off eight pawns suffice: `W a2 a4 c2 e2 / B a3 a6 d7 f7` is the candidate, moved two files over, beside an a-file group worth $\star$ [C2]. Rings give $\star 3$ in a single component. Outside the default setting it comes early: with e.p. off, the eight-pawn ring `W a5 b3 c2 d2 / B a7 b5 c6 d4` on C_4 under the opponent rule, and the ten-pawn ring `W a2 a3 b2 b3 c3 / B a6 a7 b6 b7 c6` on C_3 under the count rule [C2]. In the default setting it appears on C_7: the same ten-pawn rings include eight of value $\star 3$, among them `W a2 a5 c4 e2 f2 / B a7 b4 d7 e5 g7`, which is $\star 3$ in both models. For all eight the scanner, our engine as a whole-board game and the third solver agree [C2]; as sums of components six keep the value. So $\star 3$, like $\star 2$, needs no more than eight ranks.

In the model, then, the $\star 2$ question has a positive answer on the ordinary 8×8 board: four files and seven pawns beside T suffice, while Elkies' higher nimbers need larger boards [6]. Whether real chess agrees is the subject of Section IX.

IX. From the Model to Real Chess

To test a model value, we keep its tree of pawn moves and replace only the leaves by real results: from the Lichess Syzygy tablebase [11] up to seven men, and beyond from tablebase-backed search and our king-and-pawn solver (Section IV).

Definition 17 (Real-chess oracle value): Let X be a pawn component on the ordinary board beside T (Fig. 3(b)). An *end position* of its model tree is a position where a line of play first reaches a passed pawn; its two *sides* are white to move and black to move. Replace each end position by $+\mathsf{B}$ if white wins it in real chess with either side to move, by $-\mathsf{B}$ if black does, by $\{ +\mathsf{B} \mid -\mathsf{B} \}$ if the side to move wins, and by 0 if the side to move loses. If a side is drawn, the *draw treatment* counts that draw as a win for black, as a win for white, or as worth 0; in the last case the end position is $\{ \mathsf{B}r_W \mid \mathsf{B}r_B \}$, where $r_W, r_B \in \{-1, 0, 1\}$ are white's results with white and with black to move. All these games lie between $-2\mathsf{B}$ and $2\mathsf{B}$. The canonical value of the resulting tree is the *oracle value* $O(X)$.

Lemma 18 (Sandwich; [P]): Replace every end position not known to be $\pm\mathsf{B}$ by $-2\mathsf{B}$, or by $+2\mathsf{B}$, and let O^- and O^+ be the resulting values. Then $O^- \le O(X) \le O^+$ whatever the draw treatment and the unknown results; if $O^- = O^+$, then $O(X)$ is certified.

Proof: Every leaf lies between $-2\mathsf{B}$ and $2\mathsf{B}$, and raising an option never lowers a game [4]; induct from the leaves. ■

A. The Textbook Candidate in Real Chess

We build the tree under either passed-pawn test, the *textbook tree* or the *e.p.-aware tree*.

Theorem 19 (Candidate in real chess; [C1]): For the candidate `W a2 c2 / B b7 d7` beside T (Fig. 7(a)):

(a) *In real chess the side to move wins; only 1… b5 wins for black, 1. c4 wins for white, and every king move loses [C2].*
(b) *In the e.p.-aware tree, under every draw treatment, O has left stop ≥ 1 and right stop ≤ -1, so O is neither 0 nor $\star 2$.*
(c) *If draws count as wins for black, $O = \{ 2 \pm 1 \mid -2 \pm 1 \}$ in both trees.*

Real chess thus agrees with $\star 2$ on the outcome and on every winning first move resolved so far (1. a4 is still open), but not on the value: in the e.p.-aware tree, which matches chess, O is always hot. In the textbook tree $\star 2$ is ruled out only when draws count as wins for one side, as in (c), so the candidate's $\star 2$ depends on the textbook test.

A second candidate, `W a3 b3 c2 / B b5 b7 c6` (10 men with T), is $\star 2$ under the opponent rule with en passant on (Section VIII); its oracle value is positive under every draw treatment [C1]. The other non-default $\star 2$ positions are open.

B. Towards a Real-Chess $1/8$

The $1/8$ position of Section VII (Fig. 6(a)) has 10 men. Its e.p.-aware tree has 302 end-position sides with at most seven men, and the tablebase resolves them all. In 290 of them only one side has a passed pawn: 283 agree with the model, 5 disagree and 2 are draws; the other 12 are races. The remaining end positions have more men. A search for the end positions that decide the value finds five with eight men and three with nine: if these go as the model says, then $O = 1/8$ whatever the rest do. Four of the eight-man positions agree with the model on both sides, except black to move in `W a3 a5 / B b4 b5`, a black win; a second search allows for that result and adds one more eight-man position, which agrees on both sides. Our king-and-pawn solver (queen promotions only) shows that the three nine-man positions agree on both sides, and that white to move in the last eight-man position, `W b2 b5 / B a5 b4`, wins. Black to move there is the last open side.

Proposition 20 (Status of $1/8$; [C1]): For the $1/8$ position, with queen promotions only in our solver, $0 \le O \le 1/8$; if black to move loses in `W b2 b5 / B a5 b4`, as the model assigns, then $O = 1/8$, whatever the other open sides do.

A real-chess $1/8$ is still conditional on the frozen kings (Section IV) and on a rerun with underpromotions. The same

TABLE IV
PAWNLESS ENDINGS AGAINST THE BARE KING ON THE ORDINARY BOARD, CYLINDER AND TORUS: SHARE WON WITH WHITE TO MOVE AND LONGEST MATE [C2]

	Ordinary		Cylinder		Torus	
Material	Won (%)	Mate	Won (%)	Mate	Won (%)	Mate
KQK	100	10	100	10	0	—
KRK	100	16	2.80	3	0	—
KBBK	99.98	19	100	14	0	—
KBNK	99.51	33	0.0245	2	0	—
KNNK	0.0107	1	0.0077	1	0	—
KRBK	100	16	99.74	13	0.0218	1
KRRK	100	7	100	8	100	7
KQRK	100	6	100	5	100	5

Mate: longest forced mate in moves; —: no position is won. KBBK has the bishops on opposite colors.

TABLE V
KING AND PAWN AGAINST KING: LEGAL POSITIONS AND WHITE WINS ON THE 8×8 ORDINARY BOARD AND CYLINDER [C2]

	White to move		Black to move	
Board	Legal	White wins	Legal	White wins
Ordinary	163,328	124,960	168,024	97,604
Cylinder	160,656	126,736	165,984	92,760

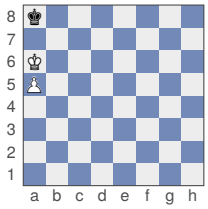


Fig. 8. Rook-pawn position Ka6, Pa5 against Ka8 (cylinder: files a and h adjacent), a white win with either side to move. On the ordinary board it is a draw with either side to move.

method applies to $\uparrow^2\star$ and $\mathrm{tiny}(1)$: their lower bounds already equal the target values, with 11 and 13 end positions of 8 to 10 men still open [C1]. Positions A and B have 11 men beside T, and the five-file $\star 2$ `W a3 b3 c2 e2 / B a6 c5 d5 e3`, which sits legally beside T, has 12, with only file f free of pawns between them. Their oracle values are open. For A and B a search finds end positions that decide the value; the five and seven of them with 8 men agree with the model (queen promotions only) [C1], but those with 9 to 11 men are unsolved, so the $\star 2$ of Theorem 16 is so far a model value.

X. MATING WITHOUT CORNERS: TABLEBASES

We now bring in the pieces and ask a classical question: without corners, which material can still force mate? We computed the tablebases of eight pawnless endings, king and one or two pieces against the bare king, on the ordinary board, on the cylinder and on the torus, where ranks 1 and 8 are joined as well. Chess on these boards is an old variant [1]. That king and rook cannot force mate on the cylinder, nor king and queen on the torus, is known in prose [12], [13], but we found no earlier tablebase for these boards.

Rules. Pieces move as in Section II and wrap round each glued edge. There is no castling and no fifty-move rule, and stalemate and bare kings are draws. A position is legal when the kings are not adjacent, across the seams included, and the side not to move is not in check. Both dimensions of the board are even, so the coloring of the squares survives the gluing.

Verification. Two retrograde solvers, written independently from these rules, agree on every position of all 24 tables, about 607 million values with either side to move [C2]. On the ordinary board both reproduce the known longest mates.

On the cylinder the queen ignores the seam: KQK wins every position, as quickly as on the ordinary board. The rook and the minor pieces do not, and on the torus not even the queen can force mate (Table IV).

KRK and KBNK cannot force mate on the cylinder. The first and eighth ranks are still edges, but now closed loops with no corner. The rook wins only 2.80% of the positions, all from mating nets already in place, such as Ka3, Rb2 against Ka1: 1. Rb3 Kh1 2. Rg3 Ka1 3. Rg1#. The rook covers h1 and the white king h2, the two flight squares across the seam. Bishop and knight win even more rarely: their mate needs a corner of the bishop's color, and the cylinder has none.

Bishops on opposite colors win every position on the cylinder, and faster than on the ordinary board, although the bishop pair is said not to mate there [13]. Here the seam helps the attacker: every diagonal winds round from the first rank to the eighth. The longest line ends in Ke6, Be1, Bf1 against Ke8: the bishop on f1 checks along f1–h3–a4–e8, across the seam, and the bishop on e1 covers d8 and f8.

On the torus only KRRK and KQRK force mate, and they win from every position. KRBK wins only where a mate in one is already set up, and the bishop pair has mating positions that no legal white move can produce.

A. King and Pawn Against King

We solved every legal KPK position on the 8×8 ordinary board and cylinder twice, with our king-and-pawn solver of Section IV (all four promotions) and with a separately written KPK solver (queen promotion only). They agree on all 326,640 cylinder positions and differ on exactly 6 ordinary-board positions, where only a rook promotion avoids stalemate, such as Ka4, Pb7 against Ka6 [C2]. Table V gives the counts; the cylinder has fewer legal positions, since the kings may not touch across the seam.

Fig. 8 shows the classic rook-pawn draw: on the ordinary board the black king cannot be driven out of the corner without stalemate. On the cylinder we may shift every man four files to the right, which gives Ke6, Pe5 against Ke8, the textbook win whoever is to move. Back on the a-file, with black to move, play goes 1… Kb8 2. Kh7 Kb7 3. a6+, and the pawn promotes under the king, which guards a7 and a8 from h7 across the seam.

Of the positions legal on both boards, 16,538 ordinary-board draws become cylinder wins. The seam also helps the defender, since a king on the h-file now stands beside an a-pawn: 15,930 ordinary-board wins become cylinder draws. Zugzwang, the engine of the trébuchet, is more common on

TABLE VI
KING AND PAWN AGAINST KING ON C_n: WHITE-TO-MOVE WINS AND MUTUAL ZUGZWANGS [C2]

n	Legal, white to move	White wins (%)	Mutual zugzwangs
3	5,706	53.00	0
4	16,008	65.87	208
5	34,350	73.92	95
6	63,036	76.55	132
7	104,370	77.49	175
8	160,656	78.89	216



(a) (b)

Fig. 9. Starting position of Dawson's chess on six files. (a) 3×6 ordinary board. (b) 3×6 cylinder: files a and f are adjacent, and the two dashed edges are glued.

the cylinder, with 216 mutual zugzwangs (white to move only draws, black to move loses) against 160 on the ordinary board [C2].

Other widths. We solved the same ending on every cylinder C_n, $n = 3, \ldots, 10$ (Table VI). Two independently written solvers agree for $n \leq 8$ [C2]; for $n = 9$ and 10 one widened solver gives 79.41% and 79.94% [C1]. Odd and even widths behave alike: the win rate rises with n, and from $n = 4$ on the share of mutual zugzwangs among white-to-move positions falls, from 1.30% to 0.13% at $n = 8$; C_3 has none, and the only even-only feature, a king on the opposite file, adds a few. The trébuchet of Fig. 3(a) is a mutual zugzwang on C_n for $4 \leq n \leq 8$ [C2] and $n = 9$ [C1], and illegal on C_3, where its kings touch [C2]. Two positions show the seam at work. Ka3, Pa2 against Ka5 is a draw with either side to move on the ordinary board, but a mutual zugzwang on every cylinder with 4 to 8 files [C2], and with 9 and 10 files [C1]. On C_3, Ka1, Pa7 against Ka3 wins at once, since a8=Q is mate. The queen checks along the a-file, which also covers a4; its wrapped diagonals cover b3, b4, c3 and c4, and the white king covers a2, b2 and c2 [C2].

XI. DAWSON'S CHESS ON THE CYLINDER

On a cylinder every file is alike, and the first move opens the ring of files into an ordinary strip. The earlier study [9] used this idea on Dawson's chess but stated the game incorrectly; here we restate it and redo the analysis.

Rules. Dawson's chess [7] is played on a $3 \times n$ board (Fig. 9) with white pawns on rank 1, black pawns on rank 3 and rank 2 empty. Pawns move and capture as in chess, captures are compulsory, and no pawn can double-step or promote. The last player to move wins; in the misère form that Dawson posed, the last player loses.

Let us see what one move does. Call a file *untouched* if both its pawns stand on their starting squares. Suppose white plays c1–c2 with the b- and d-files untouched. The pawn on c2 attacks b3 and d3, so black must capture, say b3×c2. That pawn attacks b1 and d1, so white must recapture, say d1×c2, and then come d3×c2 and b1×c2. This leaves black to move, with the b- and d-files empty and the c-file blocked for good, since a pawn enters a file only by capturing. At the end of a row the same push takes three plies and empties its one neighbor; a push with no untouched neighbor is simply blocked.

Each such sequence is forced and touches only the pushed file and its neighbors. Counting a whole sequence as one move, strips separated by empty files never interact, and both players have the same sequences, so this game of sequences is impartial. A strip of n untouched files is therefore the octal game $\mathbf{0.137}$ [3], [14], in which a move removes a file and its untouched neighbors. Its Grundy values are $g(0) = 0$, $g(1) = 1$ and, for $n \geq 2$,

$$g(n) = \operatorname{mex}\big(\{g(n-2)\} \cup \{g(a) \oplus g(b) : a+b = n-3\}\big), \quad (1)$$

where $a, b \geq 0$, $\oplus$ is the nim-sum and mex is the least non-negative integer not in the set. The first player wins the ordinary $3 \times n$ board exactly when $g(n) \neq 0$. The values from (1) have period 34 with last exception $n = 51$ [14], which two programs reproduce for $n \leq 300$ [C2].

What was wrong. In [9] a pawn could be removed from a single row only when both its neighbors were empty, so a full row of two or more pawns has no move. Its recurrence lacks the end-file term $g(n-2)$ of (1) and the value $g(1) = 1$, agrees with g at only 31 of the 301 values $n \leq 300$, and gives the wrong cylinder outcome for 56 of them [C2]. The first is $n = 4$, a predicted first-player win: white's push and four captures leave one untouched file, black pushes that pawn, it is blocked, and white has no move.

Theorem 21 (Ring rule; [P], machine-checked for $n \leq 22$): On the full $3 \times n$ cylinder, $n \geq 3$, every first move with its forced captures leaves one strip of $n - 3$ untouched files, with the opponent to move. Hence the cylinder has Grundy value $\operatorname{mex}\{g(n-3)\}$*, and the first player wins if and only if $g(n-3) = 0$, that is, for $n = 3, 7, 11, 17, 23, 27, 31, 37, 41, 45, 57, \ldots$*

Proof: Every file of a full ring has two distinct untouched neighbors, so every push starts the five-ply sequence above. The other $n-3$ files, bounded by empty files that never change, form an ordinary strip. By rotation all first moves are alike, and the cylinder has one option. ■

The ring formula of [9] and its remark that the ring's value is always 0 or 1 are therefore right once the correct g is used (Fig. 10).

Misère play. A forced sequence has odd length and ends with the player who began it, so the ring rule carries over: the first player wins the misère cylinder exactly when the strip of $n - 3$ files is a misère P-position. Our implementation of the Plambeck–Siegel algorithm [15] reproduces their partial quotient for strips of at most 32 files (order 638, 109 P-positions) and agrees with exhaustive search on 7.4 million positions [C2]; our computation of the next quotient, for longer strips, did not finish.

A warning about sums. Valued on its own as a partizan game, the ordinary 3×5 board is $\star$, although $g(5) = 3$. The

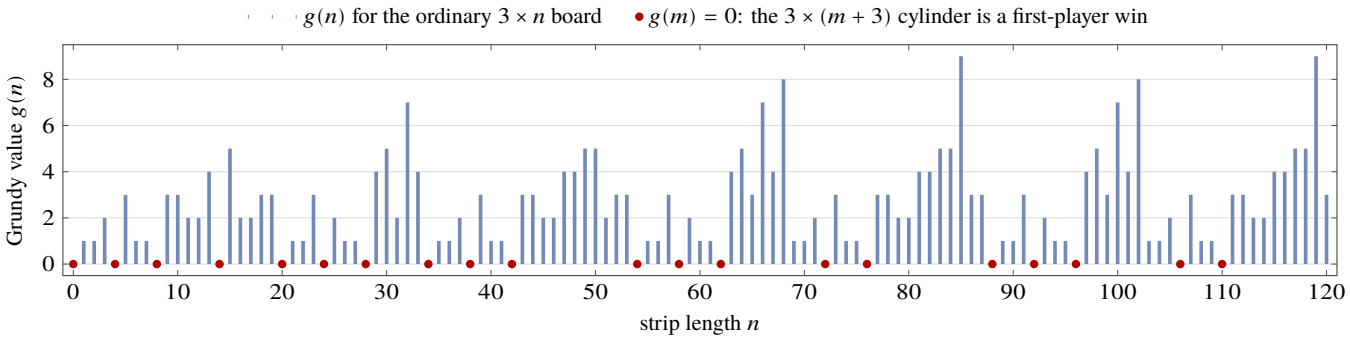


Fig. 10. Grundy values $g(n)$ of Dawson's chess on the $3 \times n$ ordinary board (octal game $\mathbf{0.137}$), $n \leq 120$. Dots mark $g(m) = 0$: the $3 \times m$ board is a second-player win and the $3 \times (m + 3)$ cylinder a first-player win (Theorem 21).

outcome agrees, but only g adds up: a sum of such values would let a player answer in another strip while a capture is pending.

XII. Engines and Learning

CGT predicts which endgames should be hard for an engine. Numbers and hot positions show in the material count or within a few moves, but an infinitesimal such as $\uparrow$ is a fight over one spare tempo and a nimber such as $\star$ a fight over parity. Neither changes the material, so a static evaluation misses them until the search reaches the end of the fight. We test this prediction on two engines.

A. A CGT Test Suite

We built 233 legal positions on the ordinary 8×8 board with at most seven men, each labeled by the class of its CGT value (Table VIII). In 181 of them the Lichess tablebase [11] confirms the winning moves that CGT predicts; in the other 52 the model and the tablebase disagree, so we score the engines against the tablebase.

B. Stockfish and Lc0

We ran Stockfish 19 [16] on one thread with a 64 MB hash and no tablebases. It solves all 233 positions at 10^5 nodes, and 232 already at 10^3 [C1]. The classes separate at a small fixed depth (Table VIII). At depth 1 the engine reads hot positions, numbers and switches correctly 73–81% of the time, but infinitesimals only 30% (26% at depth 2) and nimbers 53% [C1]: exactly the tempo and parity fights that CGT singles out. Both classes reach 100% by depth 8. With one engine and a small sample, this is a pilot result.

Lc0 v0.32.1 [17], with its bundled network, is perfect at all four budgets, 10^3 to 10^6 nodes: every best move keeps the tablebase result and every evaluation matches it, and it calls all 17 draws correctly from 10^3 nodes on [C1]. At practical budgets the suite is therefore too easy for both engines; only Stockfish at small fixed depth separates the classes.

C. ChessSums

Beyond seven men no tablebase applies, but CGT still values sums exactly. We built ChessSums, a set of 26,000 pawn-endgame sums whose components come from a library of 418 with 37 distinct values; every component has the same value in all nine settings (Section II-D). Whole-board alpha–beta confirms every outcome except five that hit its time limit [C1], and an independent minimax confirms 4,887 of them [C2]. Alpha–beta, blind to the split, needs about 4–9 times as many nodes per added component [C1], while adding the CGT values costs almost nothing.

D. Engines on Sums

We placed each sum on the ordinary 8×8 board beside T (Fig. 3(b)) or a shifted copy that our king-and-pawn solver confirms is a mutual zugzwang, giving legal positions with 15–20 men. On 200 of them, Stockfish 19 picks a move in the winning set only 26.5–28.0% of the time from 10^3 to 10^6 nodes, and Lc0 32.5–33.5% at 10^3 and 10^4 nodes [C1] (45 of the 200 are second-player wins with no winning move; excluding them, 34–36% and 42–43%). More nodes barely help, both engines get worse as components are added, and the errors do not single out nimbers or infinitesimals.

The ground truth here is the CGT model with frozen kings, not real chess, and frozen kings are not always exact (Section IV-D). These numbers therefore measure agreement with the model. Proving real-chess errors would need the kind of certification that Section IX begins for $1/8$.

E. Learning Dawson's Chess

Can a network learn CGT? In normal-play Dawson's chess (Section XI) the first player wins exactly when the strips' Grundy values have a nonzero nim-sum [3], [14]. An LSTM value network in PyTorch [18], trained on total lengths up to 30, reaches 0.82 test accuracy on total lengths 31–80, against about 0.79 for the majority guess [C1]. Linear probes on its strip embedding do no better than the majority guess, either for a strip's Grundy value (0.06 against 0.24) or for whether a single strip is a first-player win (0.84, the share of such strips). The rule itself scores 1.0, so the network has not learned it, and the probes find no sign of it either.

TABLE VII
MAIN POSITIONS WITH KINGS

Role and value	Board	Men (white / black)	FEN
Kings and puzzles			
Trébuchet across the seam: mutual zugzwang	C_8	`W Kh5 a4 / B Kb4 a5`	`8/8/8/p6K/Pk6/8/8/8`
Trébuchet T: mutual zugzwang	F_8	`W Kf5 g4 / B Kh4 g5`	`8/8/8/5Kp1/6Pk/8/8/8`
King group of Puzzle 2: white to move draws	C_8	`W Ke6 d7 / B Kd8 e7`	`3k4/3Pp3/4K3/8/8/8/8/8`
Puzzle 2: pawn part an infinitesimal of mean 0	C_8	`W Ke6 a2 b2 d7 h3 / B Kd8 a6 b7 e7 h7`	`3k4/1p1Pp2p/p3K3/8/8/7P/PP6/8`
Puzzle 3 as drawn: +B; illegal (white pawns on rank 1)	C_8	`W Kg8 a1 a5 c1 c3 d3 f7 g1 h1 / B Kf6 c2 d4 d5 d6 g3 g7 h4`	`6K1/5Pp1/3p1k2/P2p4/3p3p/2PP2p1/2p5/P1P3PP`
Elkies' small values			
$1/8$, in all nine settings	F_8+T	`W a2 b2 b3 / B a5 a6 b6`	`8/8/pp6/p4Kp1/6Pk/1P6/PP6/8`
$\uparrow^2\star$, in all nine settings	F_8+T	`W a2 b2 b6 / B a6 b7 c5`	`8/1p6/pP6/2p2Kp1/6Pk/8/PP6/8`
Tiny 1	F_8+T	`W a2 b2 b6 / B a6 b7 c6`	`8/1p6/pPp5/5Kp1/6Pk/8/PP6/8`
Textbook $1/8$, now $-3/2 \pm 3/2$	F_8+T	`W a2 b2 d2 / B a5 c4 d7`	`8/3p4/8/p4Kp1/2p3Pk/8/PP1P4/8`
Textbook $\uparrow^2$, now $\{\, 3/4 \mid \downarrow\star \,\}$	F_8+T	`W a3 c5 d4 / B a7 b7 d7`	`8/pp1p4/8/2P2Kp1/3P2Pk/P7/8/8`
The $\star 2$ question			
The candidate: $\{\, 2 \pm 1 \mid -2 \pm 1 \,\}$; $\star 2$ under the textbook test	F_8+T	`W a2 c2 / B b7 d7`	`8/1p1p4/8/5Kp1/6Pk/8/P1P5/8`
Second candidate: $\star 2$ under the opponent rule; oracle value positive	F_8+T	`W a3 b3 c2 / B b5 b7 c6`	`8/1p6/2p5/1p3Kp1/6Pk/PP6/2P5/8`
Position A: $\star 2$	F_8+T	`W a4 b3 d2 d3 / B a7 b7 c5`	`8/pp6/8/2p2Kp1/P5Pk/1P1P4/3P4/8`
Position B: $\star 2$	F_8+T	`W b4 c2 d2 / B a6 a7 c6 d5`	`8/p7/p1p5/3p1Kp1/1P4Pk/8/2PP4/8`
Five-file $\star 2$	F_8+T	`W a3 b3 c2 e2 / B a6 c5 d5 e3`	`8/8/p7/2pp1Kp1/6Pk/PP2p3/2P1P3/8`
Real chess			
Last decisive end position for $1/8$: white to move wins, black to move open	F_8+T	`W b2 b5 / B a5 b4`	`8/8/8/pP3Kp1/1p4Pk/8/1P6/8`
End position where model and chess disagree: black to move, black wins	F_8+T	`W a3 a5 / B b4 b5`	`8/8/8/Pp3Kp1/1p4Pk/P7/8/8`
KPK rook pawn, either side to move: a cylinder win, an ordinary-board draw	C_8	`W Ka6 a5 / B Ka8`	`k7/8/K7/P7/8/8/8/8`

TABLE VIII
STOCKFISH 19 AT FIXED DEPTH: SHARE OF EVALUATIONS (%) THAT MATCH THE TABLEBASE, BY CGT CLASS. SEVEN POSITIONS ALREADY DECIDED (VALUE +B OR −B) ARE OMITTED

		Search depth					
CGT class	n	1	2	4	6	8	12
hot (±B switch)	68	81	78	91	97	99	100
number	57	79	81	84	93	95	98
switch	37	73	78	89	100	100	100
other	20	60	50	75	95	100	100
nimber	17	53	65	88	100	100	100
infinitesimal	27	30	26	59	85	100	100

Infinitesimal: ups and downs, with or without $\star$. Other: values free of B that fall in none of the other classes, such as tinies and games $\{\, a \pm x \mid b \pm y \,\}$.

XIII. VERIFICATION AND REPRODUCIBILITY

Cross-checks. Our CGT engine agrees with a separate brute-force sum solver on $55{,}195$ positions G (at most two pawns a side, 3–5 files), for three race rules and 15 reference games X: all $4{,}967{,}550$ outcomes of $G + X$ match under the textbook test, e.p. on and off, and all $2{,}483{,}775$ under the e.p.-aware test. A whole-board minimax confirms its outcomes on $1{,}070$ single components. Our legal-move generator matches Lichess on all 160 test positions, covering en passant, promotions, pins, checks, mates and stalemates. Each section states its other cross-checks. The AI systems used, and what each did, are listed in the Acknowledgment.

Code availability. The code and data are available in an anonymized repository supplied to the editor. Our CGT engine and most tools are in Python 3 (standard library only). The fast solvers (the king-and-pawn solver, the census scanner, the tablebase generators) are in C. Real-chess checks use the Lichess Syzygy tablebase API. The engine experiments use Stockfish 19 and Lc0. The learning baselines use PyTorch.

XIV. CONCLUSION AND OPEN PROBLEMS

Elkies' sum model holds up on a board without edges once the passed-pawn test respects en passant and the frozen kings are checked. The textbook $\star 2$ comes from the textbook passed-pawn test, and $\star 2$ appears with seven pawns. On the cylinder, king and rook can no longer force mate, and the rook-pawn draw Ka6, Pa5 against Ka8 becomes a win. On the $3 \times n$ cylinder, Dawson's chess is a first-player win exactly when the strip three files shorter is a second-player win. Stockfish at small fixed depth loses most on the tempo and parity fights that CGT singles out, and on sums of 15 to 20 men both engines often miss the model's winning moves. Seven problems are open.

1) *A certified* $1/8$. One decisive side of the $1/8$ remains, black to move in `W b2 b5 / B a5 b4`. If it is a white win, as the model predicts, and the queen-only results survive a rerun with all four promotions, the

oracle value is exactly $1/8$ (with the kings frozen inside the tree). The same pipeline applies to $\uparrow^2\star$ and Tiny 1 (Section IX-B).

2) *The seven-pawn $\star 2$ in real chess.* Positions A and B have open oracle values: the 8-man end positions that decide them agree with the model, and those with 9 to 11 men are unsolved (Section IX-B). That A and B are the only seven-pawn $\star 2$ positions of width at most four rests on one program [C1].
3) *Higher nimbers.* Does the first ring $\star 2$ on C_n keep having $n + 3$ pawns (eleven on C_8)? In the default setting, is there a $\star 3$ with fewer than nine pawns, or a $\star 4$ on eight ranks?
4) *When is a trébuchet zero?* Prove Conjecture 5, or settle Conjecture 7; real trébuchets seem to have h of about 5 to 7 [Conj].
5) *Endgames without edges.* KPKP and KPPK on the cylinder come next. On the Möbius strip, where files a and h are joined with the ranks reversed, KPK is won from 78.94% of the legal positions with white to move, against 78.89% on the cylinder [C2]. Pawn play across its twisted seam, where a pawn reverses direction, and the Klein bottle remain open.
6) *Misère Dawson's chess.* Dawson's own question was about misère play [7], and our computation of the quotient after that of Plambeck and Siegel [15] did not finish: is the misère quotient of Dawson's chess infinite? By the ring rule (Theorem 21), any answer for strips also answers the cylinder.
7) *Engines beyond the tablebases.* Are the engines' misses on sums of 15 to 20 men errors in real chess? No tablebase judges those positions; certified components beyond seven men would give ground truth.

APPENDIX
POSITIONS

Table VII gives the FEN of each main position with kings. A men list gives the white men, then the black men; K marks a king. On "F_8+T" the position stands on the ordinary 8×8 board beside T (Fig. 3(b)), and its FEN includes T. FENs give the piece placement only; white is to move unless stated, and values are in the default setting (Section II-D). Positions with pawns only are given by their pawn lists in the text; their FENs follow directly.

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The author thanks the maintainers of the Lichess tablebase, Stockfish and Lc0 for making their tools freely available. AI systems were used throughout, under the author's direction; the author is responsible for the content. Claude (Opus 5.5, Anthropic), in Claude Code, wrote most of the software, ran the searches, drafted the proofs, figures and text from the 2025 study [9] and the notes, and directed Claude Sonnet 5.5 and Haiku 5.5 agents that audited the proofs and checked results. Codex (OpenAI) wrote the independent whole-board solver, the census scanner and the third ring solver. Nemotron 3 Ultra (NVIDIA), MiniMax M3.1 Flash and unnamed router models, run through Kilo, OpenCode and OmniRoute, repaired proofs, recomputed king-and-pawn numbers, ran delegated checks and the experiments of Section XII. Statements labeled [AI] have been checked only by these systems.

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